

Neural Population Manifolds in flyvis

Johannes Bertram

Abstract

This work applies the recently developed neural encoding manifold technique to the flyvis artificial neural network model of the fly visual system. The goal is to determine whether flyvis shows smooth manifolds that were found in mouse V1 but not in artificial neural networks. The manifolds show clear clustering of cell types with increasing spread across the processing stages in flyvis. Excitatory and inhibitory cells cannot be clearly distinguished in the manifolds. The manifold in the last processing stage exhibits the most flexible behavior. The manifold is validated by finding activation patterns of T4a and T4b neurons consistent with the known motion detection function. Unlike the manifolds found in mouse V1, the flyvis manifolds are not smooth highlighting significant differences in processing between flyvis and mouse cortex.

1 Introduction

Recent advances in fruit fly connectome research allowed the building of a connectome-constrained artificial neural network model of the fruit fly vision system (flyvis) [LTP⁺24]. They show that neural activity in this connectome-constrained artificial neural network aligns with measurements of experimental activity in several experiments. Therefore, flyvis is a recent attempt to approximate neural behavior in the visual cortex using a differentiable artificial neural network architecture.

As a new analysis method for neural behavior in the visual domain [DRH⁺24] develop neural encoding manifolds. They show that in the mouse visual cortex, neurons lie on a smooth manifold in a stimulus-response space. This means all neurons exhibit slightly different temporal response patterns to various stimuli. Opposed to that, the neurons in the mouse retina are grouped in clusters that have similar response patterns within. Furthermore, they find that artificial neural networks do not exhibit smooth representation but instead are heavily clustered.

In this work, I will build neural encoding manifolds at different processing stages in the fly visual system and examine their smoothness.

2 Method

In the following, the flyvis architecture, the transferability of the approach, and the sampling procedure are discussed. For the procedure of building the neural encoding manifold and their stimuli, refer to [DRH⁺24].

2.1 flyvis

Based on experimental measurements of connectivity in the *Drosophila* visual system (fruit fly), an equivalently connected artificial neural network is set up [LTP⁺24]. The resulting recurrent network architecture is displayed in Figure 1. Every cell type is treated as a hexagonal feature map.

This approach of using the connectome reduces the number of trainable parameters in the network. Only single-neuron and single-synapse properties are unknown and thus optimized using backpropagation. It is assumed that neurons of the same cell type share their neuron and synapse properties. These assumptions result in a reduction of trainable parameters from above 1 million to only 743.

To obtain training signals, the network is trained to perform an optical flow task. To do so, a decoder is applied to the neuron activities and predicts optical flow. Individual neurons are modeled with passive leaky linear non-spiking voltage dynamics and linear thresholding for spiking behavior.

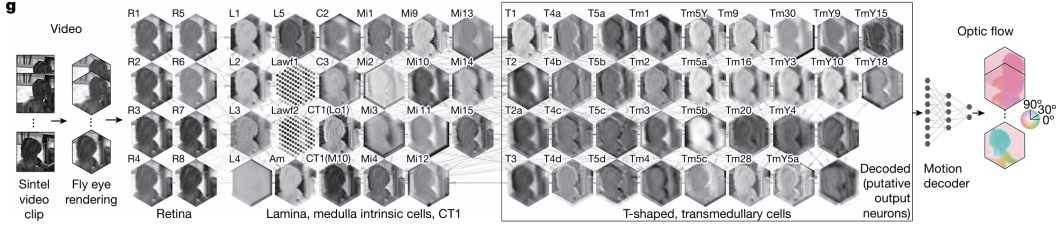


Figure 1: Cell types and architecture of flyvis. Figure taken from [LTP+24]

2.2 Transferability of Approach

In [DRH+24], the manifolds are built for mouse data using custom stimuli. Their stimuli are chosen to be both controllable and relevant to the mouse by using 6 stimulus types representing grating and dot flows in 8 flow directions. This simulated optical flow. Therefore, as flyvis is trained to perform optical flow tasks, the stimuli should also be of relevance to the fly. I argue that even though the generalization performance of flyvis to these unknown stimuli is unknown, it is valid to use these optical flow stimuli simpler than the flyvis training data. As the resolution of inputs to flyvis is relatively low, only a smaller region of the input stimuli is used and transformed to the hexagonal input grid. The exact resizing is determined using visual inspection of the resulting hexagonal input.

2.3 Sampling

The different processing stages in the mouse visual system are treated separately in [DRH+24]. Therefore, I split the network into 4 processing stages from which the activity data is sampled independently. The 4 stages are *Retina*, *Lamina*, *Medulla Intrinsic*, *T-shaped* and *Transmedullary*. Each cell type within these stages is treated as one feature map. Then, from each cell type, 50 neurons are sampled from the best-performing flyvis model.

3 Results

In the following, the neural encoding manifolds for the 4 stages are plotted. The points are colored by cell types, and for selected cell types, the mean activations for all stimuli are plotted. The activations are plotted by cell type instead of individually, as the neurons were largely clustered by cell type. The color of the titles is determined by the cell types forming either predominantly inhibitory (blue) or excitatory (red) connections with subsequent cells. Information about inhibitory and excitatory connections is taken from Fig. 1 in [LTP+24].

3.1 Stage 1: Retina

Here (Figure 2), the optimal number of factors found in the tensor decomposition is very low at 5. Increasing the number of factors did not result in improved explained variance. The resulting manifold consists of one cluster containing 7 out of the 8 retinal cell types. All of those seven exhibit high activity in response to all stimuli. R7, the cell type not in that cluster, is more spread out and exhibits various orientation-selective temporal response patterns. Due to the high variability of this cell type, additional single neuron response patterns are included in the plot.

3.2 Stage 2: Lamina

Compared to the *Retina*, the cell types in the *Lamina* are more spread out (Figure 3). All cell types can clearly be distinguished in the manifold. In the top right portion of the manifold, cell types show high responses to all stimuli. The other cell types also exhibit generally high responses but show more temporal and orientation differences. There is no clear structure in the way inhibitory and excitatory cell types are distributed across the manifold.

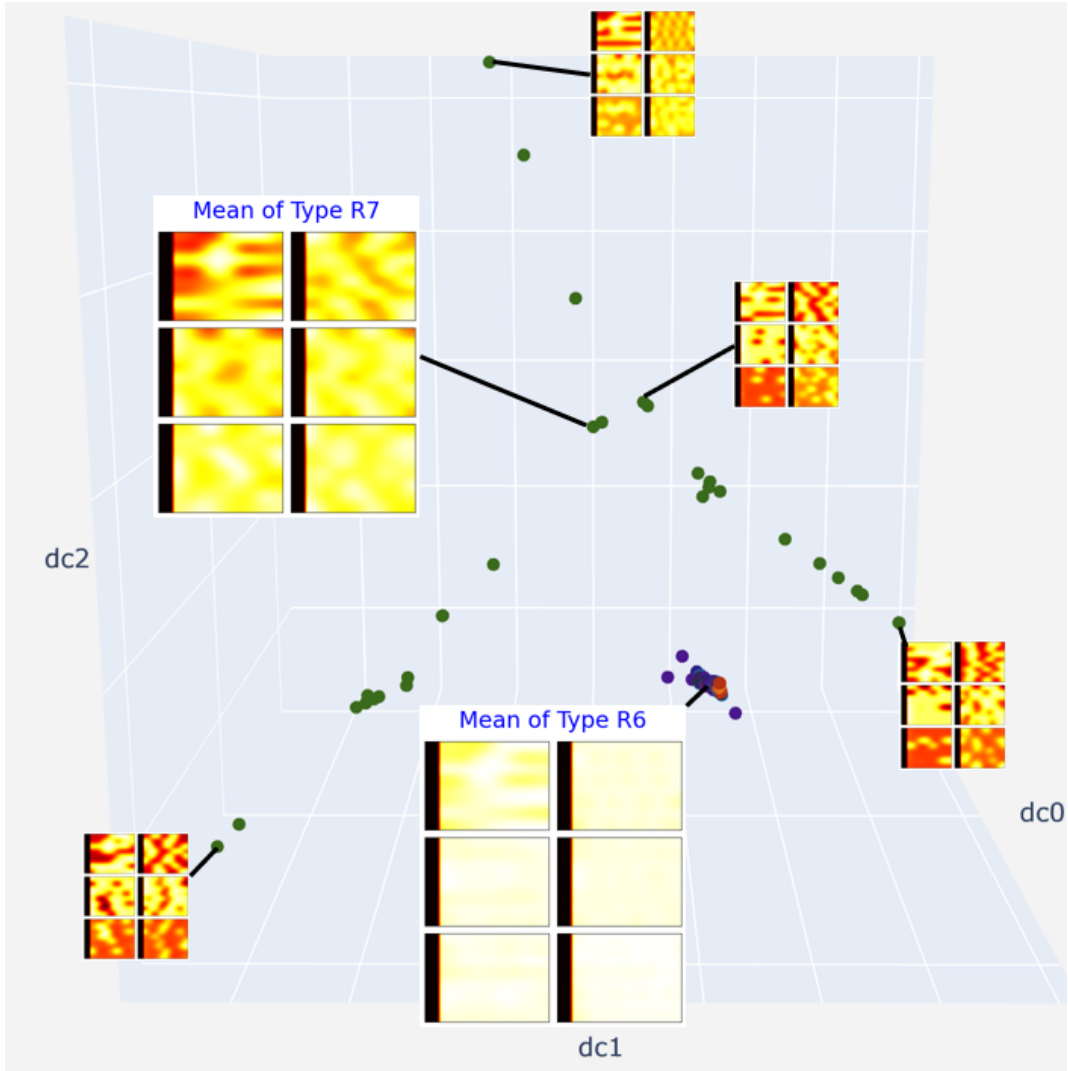


Figure 2: Neural encoding manifold of the *Retina*. Smaller subplots are additional single neuron activities

3.3 Stage 3: Medulla Intrinsic Cells

Generally, the manifold of *Medulla Intrinsic* cells (Figure 4) is very similar to the manifold of the *Lamina* (Figure 3). However, inhibitory cells tend to be located on the right side of the manifold. While in (Figure 4) more cell types are excluded to avoid cluttering, this trend remains but is not fully consistent when looking at all cell types. Following this trend, inhibitory cells tend to be more active to all stimuli, while, for example, excitatory cell Mi15 shows different temporal behaviors. Also, it is interesting to note that the cell type Mi11 is split into two clusters, both of which are very highly activated for all inputs.

3.4 Stage 4: T-shaped and Transmedullary Cells

In this manifold of the final layers, the clusters representing the individual cell types are more spread out (Figure 5). Still, the individual cell types are mostly distinguishable from others and clustered. There are two highly concentrated groups of neurons, including different cell types. First, on the right, there is a group of neurons all exhibiting high responses to all inputs.

Second, more in the center of the plot and including cell types T4a and T4b, there is a group of neurons that show generally low activity and are highly orientation-tuned irrespective of stimulus. Furthermore, these cells respond similarly to all stimuli. This is exactly the activation pattern needed

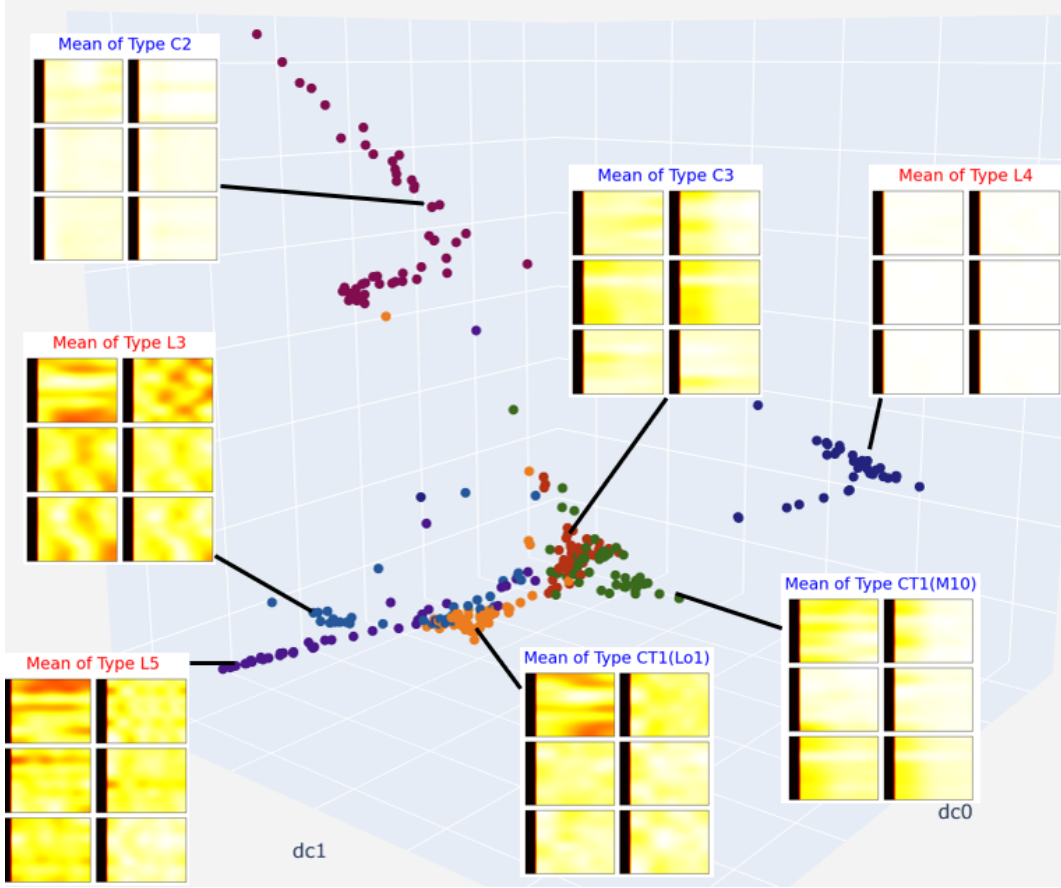


Figure 3: Neural encoding manifold of the *Lamina*

to detect a flow: Irrespective of the stimulus, the neurons are activated for one orientation and not the opposite direction, which does not differ in images but only in flow direction. This finding is supported by the fact that the cell types T4a and T4b are well-studied motion detector neurons [LTP⁺24].

In the remaining regions of the manifold, various activity patterns, including stimulus selectivity, orientation selectivity and different temporal patterns are present. These different activity patterns are more diverse than in earlier layers.

4 Discussion

A first point of discussion is that it is unclear if the chosen stimuli are adequate for the flyvis network. The generalization performance to these stimuli is unknown. Also, a significant number of cells are always highly activated irrespective of stimulus and orientation or are at least not selective for the different stimuli apart from the low-frequency grating. This might indicate that the stimuli are in a regime that is not well-captured by these neurons.

However, since later layers show different types of temporal, orientation-selective and stimulus-selective behavior and the activation even provides a way in which the neurons T4a and T4b could perform their studied task of motion detection, the stimuli appear to be adequate for at least a subpopulation of neurons. Therefore, the manifolds should also capture relevant information about the network.

Opposed to [DRH⁺24], no clear distinction between inhibitory and non-selective, opposed to excitatory and selective neurons, is present in the flyvis manifolds.

In general, the flyvis manifolds are not as smooth as the V1 manifolds in the mouse. Moving further along the processing in the visual system shows an increasing spread of neurons within their clusters. This could be analogous to the V1 in the mouse having smooth, that is, more spread out manifolds

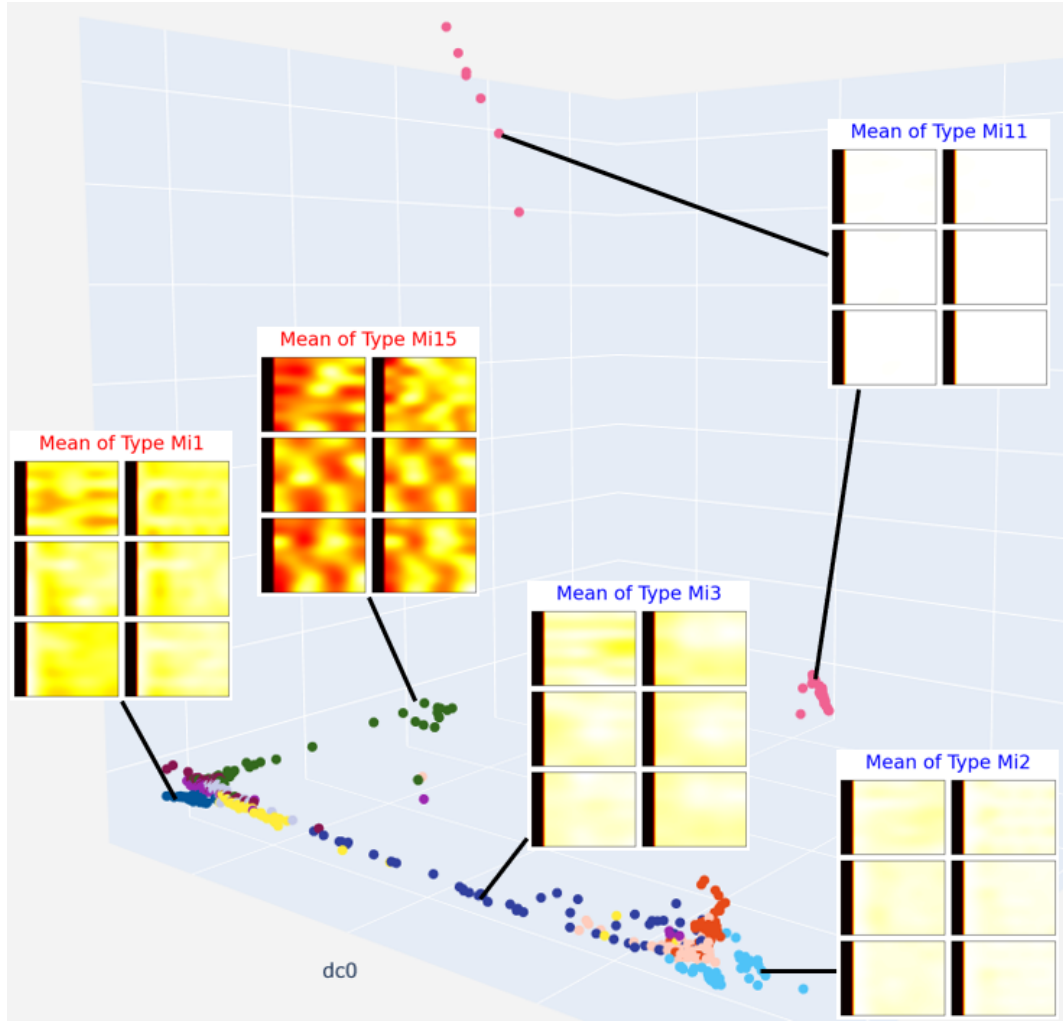


Figure 4: Neural encoding manifold of the *Medulla Intrinsic* cells

compared to the earlier stage in the retina. One could hypothesize that increasing feature complexity as obtained in deeper layers leads to more spread-out manifolds. However, there likely seems to be an additional step required to obtain smooth instead of spread-out but clustered manifolds. There is still a clear difference between the non-smooth manifold of the final flyvis processing stage and the mouse V1. Unfortunately, it is unclear whether this stems from flyvis being an imperfect model of the fly visual system or from the difference between the fly and mouse visual systems. For answering this question, it would be interesting to build neural encoding manifolds for biological measurements of the fly visual system.

References

- [DRH⁺24] Luciano Dyballa, Andra M. Rudzite, Mahmood S. Hoseini, Mishek Thapa, Michael P. Stryker, Greg D. Field, and Steven W. Zucker. Population encoding of stimulus features along the visual hierarchy. *Proceedings of the National Academy of Sciences*, 121(4):e2317773121, 2024.
- [LTP⁺24] J. K. Lappalainen, F. D. Tschopp, S. Prakhya, et al. Connectome-constrained networks predict neural activity across the fly visual system. *Nature*, 634:1132–1140, 2024.

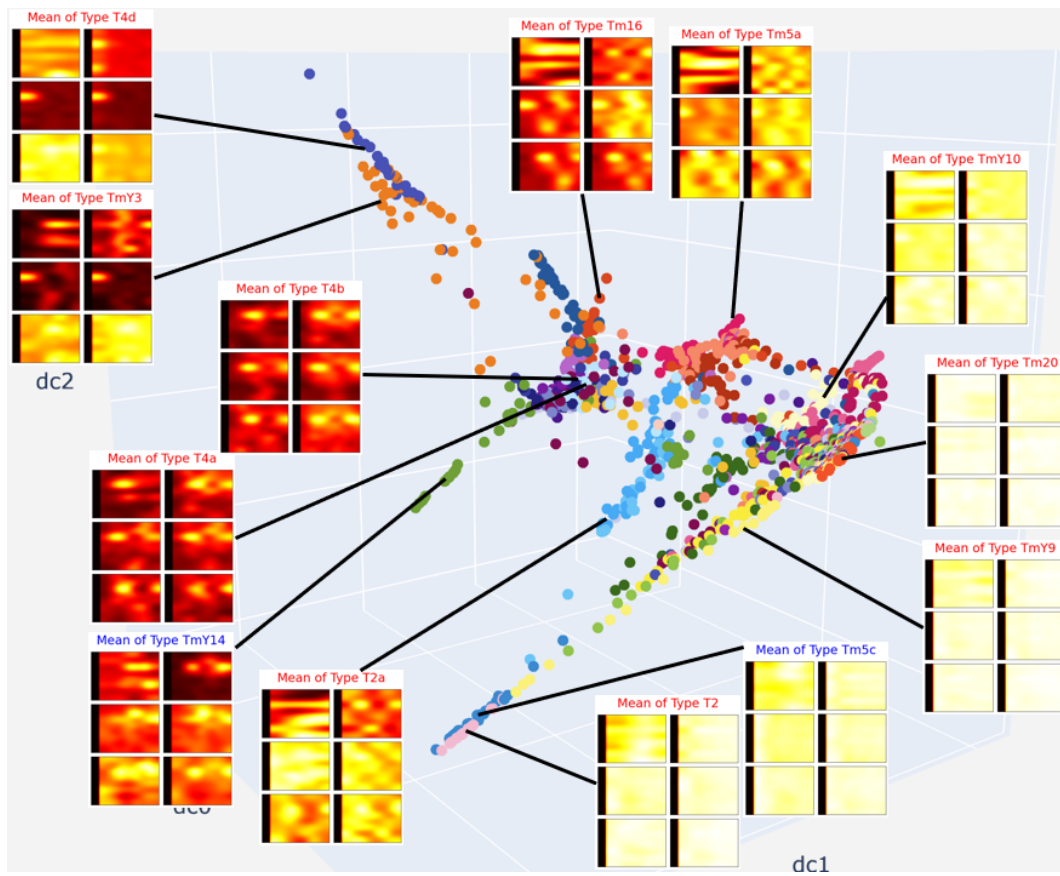


Figure 5: Neural encoding manifold of the *T-shaped* and *Transmedullary* cells